

DEDEKIND

The Programming Language for Science

Physical Units • Differentiable Solvers • GPU Native

Mario Michael Heinrich

Founder | github.com/Engineer1080 | v1.3.1

Nature's Laws Native in Code

Replacing fragile scripts with a language that understands the
fundamental grammar of physics and mathematics.

SCIENCE'S SILENT CRISIS

⚠️ Dimensional Errors

~30% of scientific software bugs stem from wrong units. Unit errors caused the \$125M Mars Climate Orbiter crash.

🔗 Boilerplate Friction

Researchers waste thousands of hours rewriting ODE solvers and manual uncertainty propagation in Python/MATLAB.

📁 Irreproducibility

Manual tracking and disconnected publication tools make scientific results impossible to verify or scale.



THE DEDEKIND SOLUTION



Type-Safe Units

Enforced at compile-time.

Adding **1[m]** to **1[s]** triggers an error instantly, not at runtime.



Differentiable

Native **grad()** support for ODE/PDE solvers, enabling effortless Physics-Informed ML and optimization.



LaTeX First

Built-in publication tools. Export verified code definitions directly to peer-reviewed LaTeX documents.

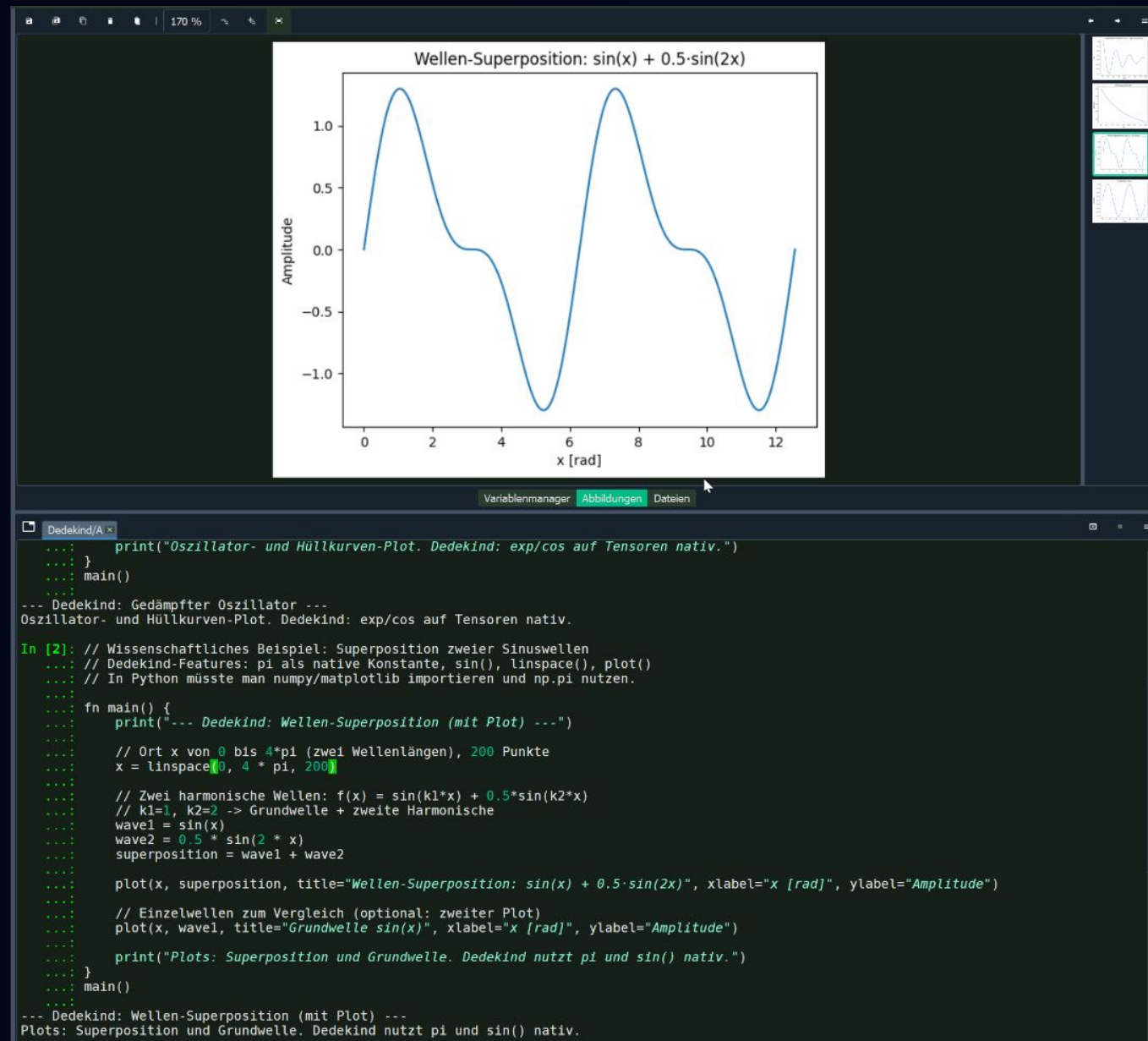
```
mass = 1.0[kg];  
energy = mass * c^2; // Automatically assigned [Joule]  
sol = ode_solve(kinetics, initial, t); // Native High-Perf Solver
```

DEDEKIND STUDIO

The Next-Gen Scientific IDE

A high-performance fork of Spyder designed for discovery.
Unified environment for coding, plotting, and real-time physical verification.

- ✓ Live Unit Verification
- ✓ Interactive Physical Plotting
- ✓ LaTeX Definition Preview
- ✓ Scientific Notation in the Terminal



UNCOMPROMISING TECH STACK

Component	Technology	Value Proposition
Compiler	MLIR / LLVM Backend	Progressive lowering for heterogeneous hardware (CPU/GPU/TPU).
GPU Acceleration	Triton Kernels (OpenAI)	Hardware-agnostic, block-level GPU programming native to the IR.
Parallelism	Work-Stealing Scheduler	Provably optimal dynamic load balancing for nested simulations.
Safety	Affine Type System	Memory safety and data-race prevention without garbage collection.

GLOBAL MARKET OPPORTUNITY

Scientific Software TAM

TAM \$28B

Numerical Tooling SAM

SAM \$6B

Early Adopter SOM

SOM \$420M

"Researchers globally waste \$2.1B annually on errors that Dedekind prevents by design."

SCALABLE REVENUE STREAMS

Open Source Core

MIT-licensed compiler and basic IDE to drive rapid academic adoption and ecosystem network effects.

SaaS & Pro

Team collaboration, cloud solvers, and premium IDE features. Recurring revenue from labs and R&D teams.

€49 - €299 / mo

Enterprise Licensing

Custom SLA, support, and embedded engine licensing for Pharma, Defense, and Engineering firms.

2026 STRATEGY & ROADMAP

Jan 2026

●
Launch

Core Unit Checker and MLIR
Prototype.

Q3 2026

●
v1.0 Release

Dedekind Studio stable for
early adopters.

Q4 2026

●
Ecosystem

100+ native solvers and
Jupyter integration.

2027

●
Scale

Enterprise cloud-native
solver launch.

FOUNDER & LEAD ARCHITECT

Heinrich Portfolio



Mario Michael Heinrich

Serial AI Builder & Language Creator

- 🏆 **4x Hackathon Winner:** Meta, XIBIX & the Max-Planck-Challenge (Anthropic x [TUM.ai](#) Hackathon).
- 💡 **Inventor:** High-performance Prime Number Generation Algorithm.
- 🚀 **Founder:** Creator of [easy-job.ai](#) and [easy-quiz.ai](#).

SEED ASK: €250,000



- 40% Full-Time Development (Compiler & IDE)
- 20% Go-to-Market (Adoption & Content)
- 20% Scientific Pilots (Uni & Pharma)
- 10% Legal & IP (Patents & SLA)
- 10% Operations & Reserve

Goal: Reach €100K ARR and 5,000 active researchers.

Questions?

Let's build the future of scientific computing together.

Resources

 github.com/Engineer1080

 Dedekind-Language.com

Contact

 Deggendorf, Germany

 MIT Licensed Core